

AI for Medicine - Chest X-Ray Classification

Updated Report: Deep Learning Classification of COVID-19, Lung Opacity, Normal, and Viral Pneumonia X-rays

Prof. Stefano Diciotti - University of Bologna

Academic Year: 2025-2026

Student Name: Reza Farjadnia

Degree Program: Electronics for intelligent systems, big-data and internet of things

Updated Report Date: 02/07/2026

1. Project Title

Chest X-Ray Classification Using Deep Learning for COVID-19 and Pulmonary Disease Detection.

2. Problem Statement

Chest X-ray images are widely used to support the assessment of respiratory conditions such as COVID-19, viral pneumonia, and lung opacity. Manual interpretation requires clinical expertise and can be time-consuming when image volume is high. This project investigates whether deep learning can classify chest X-rays into four diagnostic categories and provide a reproducible experimental workflow suitable for educational research.

The model is not intended to replace clinicians. Its value is in demonstrating how data quality controls, leakage prevention, model comparison, and explainability can be combined in a medical imaging pipeline.

3. Objective of the Study

The objective is to build and evaluate a chest X-ray classification pipeline with two stages: a custom CNN baseline and a MobileNet transfer-learning model. Model selection is performed only on the development set using stratified 5-fold cross-validation. The independent 20% hold-out test set is used once for final evaluation of the selected model.

4. Dataset Description

The project uses the COVID-19 Radiography Database from Kaggle. After quality checks and duplicate removal, the clean dataset contains 21,111 images across four classes. Each image has a corresponding segmentation mask supplied by the dataset, and mask-image consistency was verified before modelling.

Class	Clean Images
Normal	10,191
Lung Opacity	6,012
COVID	3,570
Viral Pneumonia	1,338
Total	21,111

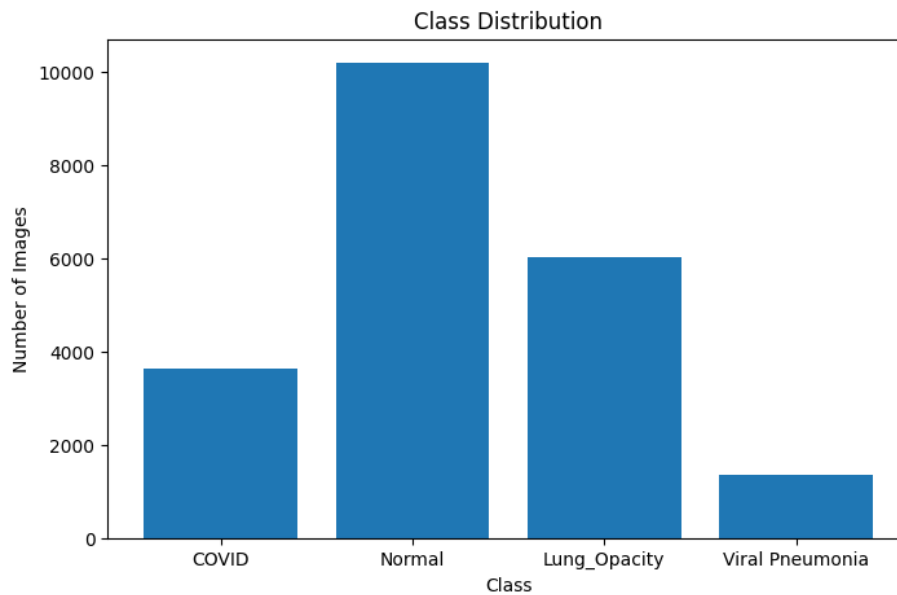


Figure 1. Class distribution after duplicate removal.



Figure 2. Example chest X-ray inspection from the cleaned dataset.

5. Data Quality and Preprocessing

The preprocessing pipeline first validates file integrity, mask availability, and duplicate images. Exact duplicate images are removed before any train-test split so the same visual content cannot appear in both development and hold-out test sets. Images are resized to 224 x 224 pixels, converted to three-channel RGB format, normalized to [0, 1], and encoded into numerical class labels.

Check	Result
Original usable image records	21,165
Clean images after duplicate removal	21,111
Exact duplicate groups	42
Duplicate images removed	54
Images with matching masks	All verified
Missing masks	0
Extra masks without matching images	0
Corrupted/unreadable images	0

No corrupted images were detected in the executed run. The mask audit found no missing masks and no extra masks without matching images. These checks improve reliability because the model is trained only on readable image-mask pairs from the cleaned dataset.

6. Avoiding Data Leakage

Data leakage was controlled by removing duplicates before splitting the data and by keeping the final hold-out test set separate from all model-selection steps. The dataset was split into 80% development data and 20% hold-out test data using stratification, preserving class proportions across splits. Cross-validation was then performed only on the development set.

This strategy is appropriate for the project because it separates model selection from final performance estimation. The cross-validation results guide the choice between CNN and MobileNet, while the hold-out test set estimates how the selected model performs on unseen data.

7. Machine Learning Pipeline

Two models were compared. The first is a custom CNN baseline trained from scratch. The second is MobileNet with transfer learning, which uses features learned from large-scale natural image data and adapts them to the chest X-ray classification task. Both models use the same cleaned dataset, labels, split strategy, and 5-fold cross-validation protocol.

Model	Mean Accuracy	Std. Accuracy	Mean Loss
Custom CNN baseline	69.43%	0.83%	0.7528
MobileNet transfer learning	88.40%	1.26%	0.3199

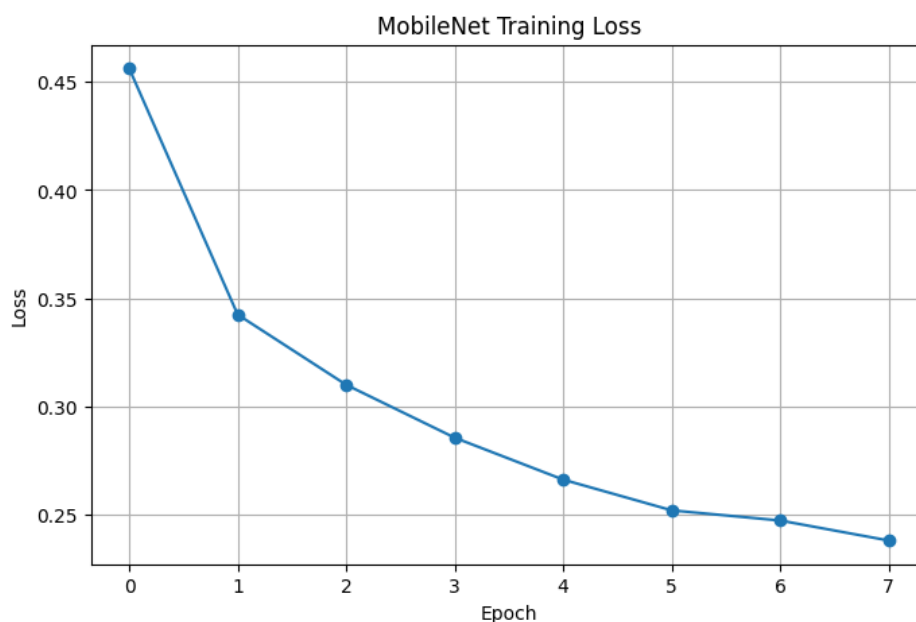


Figure 3. Training loss curve for the final MobileNet training run.

8. Results

MobileNet clearly outperformed the custom CNN baseline during 5-fold cross-validation. The CNN reached a mean validation accuracy of 69.43%, while MobileNet reached 88.40%. MobileNet was therefore selected as the winning development-set model and evaluated on the untouched hold-out test set.

Metric	Value
Selected model	MobileNet transfer learning
Hold-out test accuracy	89.72%
Hold-out test loss	0.2679
Macro-average AUC	0.9836
Weighted F1-score	0.8963

On the hold-out test set of 4,223 images, MobileNet achieved 89.72% accuracy and a macro-average AUC of 0.9836. The high AUC values show strong class separability, while the class-level report gives a more detailed view of where the model performs best and where it still needs improvement.

Class	Precision	Recall	F1-score	Support
COVID	0.8935	0.8810	0.8872	714
Lung Opacity	0.9166	0.8130	0.8617	1,203
Normal	0.8839	0.9450	0.9134	2,038
Viral Pneumonia	0.9377	0.9552	0.9464	268
Macro avg	0.9079	0.8985	0.9022	4,223
Weighted avg	0.8982	0.8972	0.8963	4,223

Class	AUC
COVID	0.9869
Lung Opacity	0.9751
Normal	0.9730
Viral Pneumonia	0.9993
Macro average	0.9836

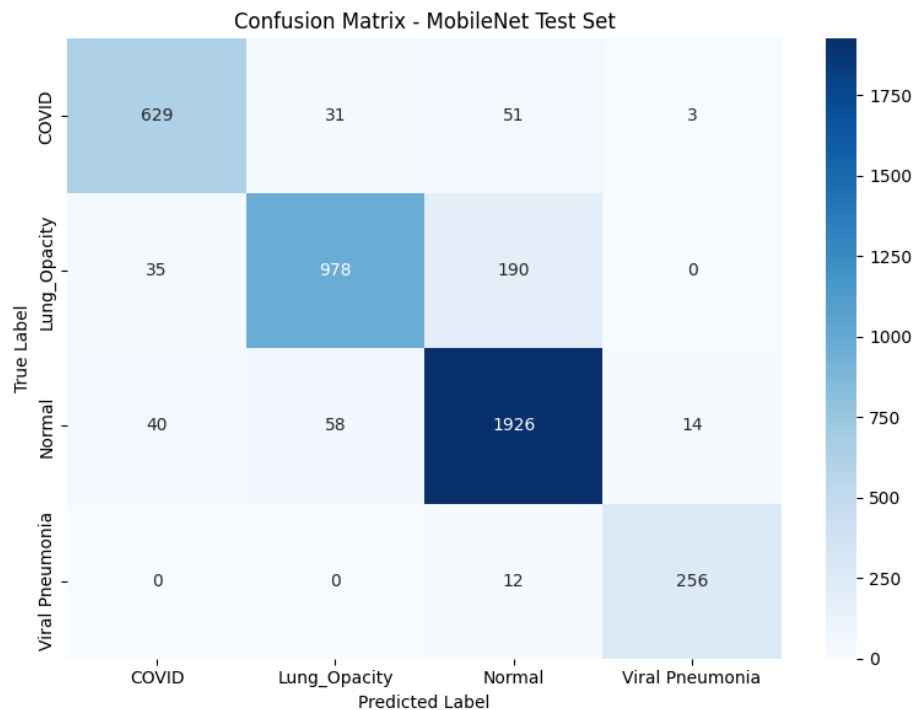


Figure 4. Hold-out test confusion matrix for the selected MobileNet model.

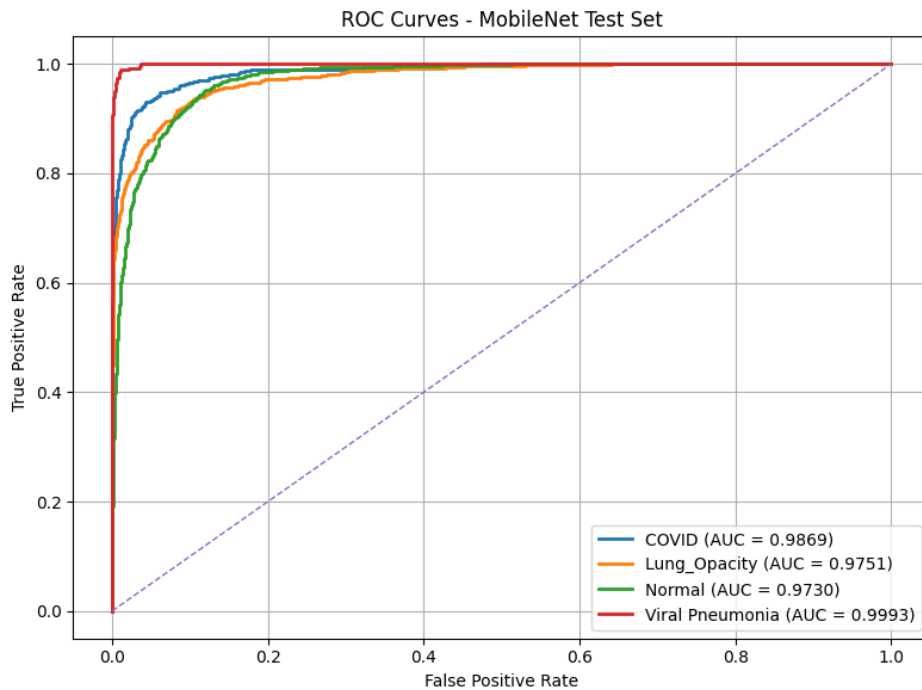


Figure 5. One-vs-rest ROC curves on the hold-out test set.

9. Model Explainability

SHAP analysis was added to improve interpretability. The latest SHAP sample intentionally includes one hold-out test image from each class: COVID, Lung Opacity, Normal, and Viral Pneumonia. The visualization highlights image regions that contribute to the predicted class. This does not prove clinical correctness, but it helps inspect whether the model is focusing on plausible radiographic regions rather than irrelevant image artifacts.

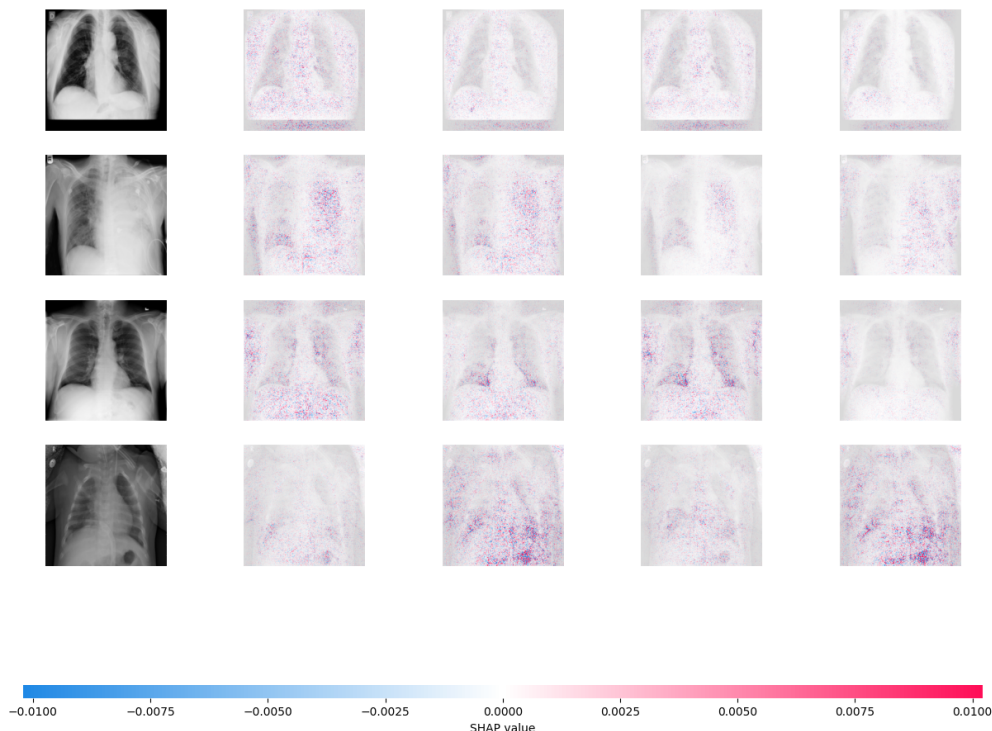


Figure 6. SHAP explanation examples for MobileNet predictions.

10. Discussion

The results confirm that transfer learning is much stronger than a small CNN trained from scratch for this dataset. MobileNet improved mean cross-validation accuracy by approximately 19.0 percentage points and generalized well to the hold-out test set.

The confusion matrix and class-level recall show that Lung Opacity remains the most difficult class. One possible reason is inconsistent image framing: some Lung Opacity images may include a wider field of view, such as head, shoulders, or hands, while other classes may be more tightly centered on the lungs. This can encourage the model to learn dataset-specific visual shortcuts instead of focusing only on lung tissue. This explanation should be treated as a hypothesis and tested directly rather than assumed.

The main weakness is Lung Opacity recall, which was 0.8130. This means the model missed more Lung Opacity cases than the other classes. In a medical imaging context, this class-level behavior matters more than accuracy alone, and future work should focus on improving sensitivity for clinically important categories.

Viral Pneumonia achieved the highest F1-score and AUC, but it also had the smallest support in the test set. Therefore, this strong performance should be interpreted carefully and ideally confirmed on an external dataset.

11. Conclusions and Future Work

The final pipeline provides a clean, reproducible workflow for chest X-ray classification. It includes data integrity checks, duplicate removal, stratified splitting, 5-fold model comparison, hold-out testing, classification reports, ROC analysis, confusion matrix analysis, and SHAP explainability. MobileNet was the best-performing model, with 89.72% hold-out accuracy and 0.9836 macro-average AUC.

Model family	Why it is useful for the next version
EfficientNetB0/B1	Often provides strong accuracy with efficient parameter usage and is a natural transfer-learning upgrade over MobileNet.
DenseNet121	Frequently performs well on chest X-ray tasks because dense connections reuse low-level and high-level radiographic features.
ResNet50V2	A robust benchmark architecture that can help verify whether the MobileNet gain is architecture-specific or general to transfer learning.
Ensemble of top transfer models	Can improve stability and class-level recall, especially for clinically important classes such as Lung Opacity.
Mask-based lung-only preprocessing	Use the verified masks to crop or hide non-lung regions, then compare Lung Opacity recall and Normal/Lung Opacity confusion against the full-image model.

Recommended next steps are to test EfficientNet, DenseNet, and ResNet variants, tune class imbalance handling, add stronger augmentation, calibrate predicted probabilities, and evaluate the final model on an external clinical dataset. A specific priority is to repeat training with lung-mask cropping or lung-only masking to check whether Normal/Lung Opacity errors are partly caused by image framing and non-lung anatomy.

12. Ethics and Data Privacy

The project uses a publicly available Kaggle dataset. No personally identifiable patient information was accessed. The work is for education and research only and should not be used as a clinical diagnostic tool without external validation, regulatory review, and clinical oversight.

13. Code and Reproducibility

The project was implemented in Python using TensorFlow/Keras, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, and SHAP. The notebook is organized so imports and dependencies are placed at the beginning, followed by reproducibility setup, data validation, preprocessing, model comparison, final evaluation, and explainability.

Experiments were designed for Google Colab with GPU acceleration. The fixed hold-out set and stratified 5-fold cross-validation protocol make the model-selection process reproducible and easier to audit.

14. References

T. Rahman et al., "Exploring the Effect of Image Enhancement Techniques on COVID-19 Detection Using Chest X-ray Images," *Computers in Biology and Medicine*, vol. 132, 2021.

Tawsifur Rahman, "COVID-19 Radiography Database," Kaggle Dataset. Available: <https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database>

A. Howard et al., "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," arXiv:1704.04861, 2017.

M. Abadi et al., "TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems," 2015. Available: <https://www.tensorflow.org>

F. Chollet et al., "Keras," 2015. Available: <https://keras.io>

F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.

S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," Advances in Neural Information Processing Systems, 2017.